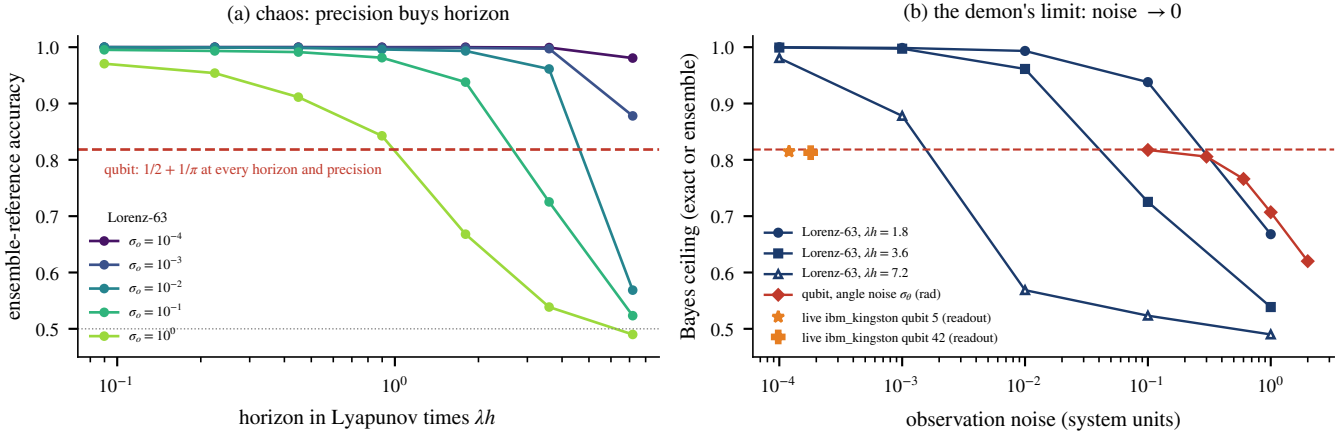
(a) chaos: precision buys horizon
ensemble-reference accuracy
1.0
0.9
0.8
0.7
0.6
0.5
qubit: 1/2 + 1/π at every horizon and precision
Lorenz-63
σ_o = 10^−4
σ_o = 10^−3
σ_o = 10^−2
σ_o = 10^−1
σ_o = 10^0
10^−1
10^0
horizon in Lyapunov times λh
(b) the demon's limit: noise → 0
Bayes ceiling (exact or ensemble)
Lorenz-63, λh = 1.8
Lorenz-63, λh = 3.6
Lorenz-63, λh = 7.2
qubit, angle noise σ_θ (rad)
live ibm_kingston qubit 5 (readout)
live ibm_kingston qubit 42 (readout)
10^−4
10^−3
10^−2
10^−1
10^0
observation noise (system units)